FC_OpenCourant – OpenCourant / OpenRadioss in FreeCAD

A short introduction

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Abstract

FC_OpenCourant (package `ORfemn`, called ORfe here) brings the explicit finite element solver OpenCourant, the continuation of OpenRadioss, into FreeCAD. It adds *OpenRadioss* as a solver of the FEM workbench, next to CalculiX, Elmer, Mystran and Z88, so an analysis set up in the usual way (mesh, material, constraints) can be solved as an explicit dynamic problem and post-processed in FreeCAD. A separate *OpenRadioss* workbench edits LS-DYNA format input decks (`.k` files) with a keyword editor and templates, and runs them. This note explains what the workbench does, how to install and use it, how it was tested, and its current limitations.

1 Why an explicit solver in FreeCAD?

The solvers bundled with FreeCAD FEM are implicit: well suited for static and modal analyses, but expensive for short, highly nonlinear events such as impacts, drop tests, crushing or forming. Explicit codes integrate the equations of motion with very small time steps and need no global stiffness matrix, which makes them robust for contact, large deformation and failure. OpenRadioss is an industry-proven explicit solver (AGPL) that reads the widely used LS-DYNA keyword format natively. ORfe connects the two: FreeCAD provides geometry, meshing and post-processing, OpenRadioss the explicit solution.

Note on OpenRadioss and OpenCourant. In October 2026 the OpenRadioss project was discontinued and its GitHub repository removed. The community continues the code as **OpenCourant** (`https://github.com/OpenCourant/OpenCourant`), with ready-to-run Linux and Windows releases. The release layout and the executable names in `exec/` are unchanged, so ORfe works with OpenCourant as it is; "OpenRadioss" in this text means either.

2 Installation

1. Install FreeCAD 1.0 or newer (tested with 1.0.0 and the 1.1.4 Flatpak).
2. Install the workbench: paste the one-line installer from `INSTALL.md` into FreeCAD's Python console (any operating system), or clone the repository into FreeCAD's `Mod` folder, then restart FreeCAD:

   ```
   git clone https://github.com/CalzitPLA/FC_OpenCourant ~/.local/share/FreeCAD/v1
       -1/Mod/FC_OpenCourant
   ```

   (FreeCAD 1.0: without `v1-1/`; Flatpak, Windows and macOS folders in `INSTALL.md`).
3. Open **Edit** → **Preferences** → **OpenRadioss** (Figure 1) and click **Download OpenCourant…**: the newest release for the platform is downloaded, unpacked, selected and tested. An existing installation is found automatically in `~/OpenCourant` or `~/OpenRadioss`, or can be selected with **Browse…**.

The page lists what a run needs: the executables, the reader configuration, the libraries and, inside the FreeCAD Flatpak, the two host libraries the solver lacks there (copied automatically). **Test installation** runs the starter on a small deck; this also catches reader configuration files that do not match the executables. OpenCourant has releases for Linux (x86-64, arm64) and Windows, not for macOS: there the deck tools work, but the solver has to run elsewhere.

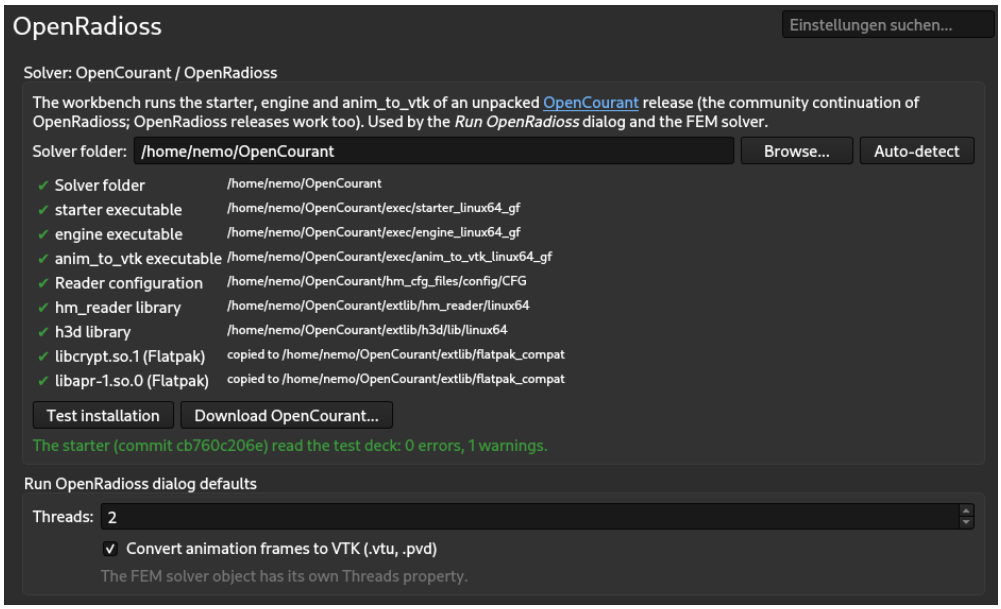


Figure 1: *Edit* → *Preferences* → *OpenRadioss* after *Test installation*.

3 How the solver is embedded: two ways in

ORfe offers two entry points, which share everything from the deck onwards (Figure 2):

FEM workbench The user sets up an ordinary FEM analysis and adds *Solver OpenRadioss*. FreeCAD's solver framework runs four tasks: *Check* (mesh, material, solver found), *Prepare* (the deck writer translates nodes, elements, materials, constraints and loads into `radioss.k`), *Solve* (starter, engine, conversion) and *Results*.

Keyword deck The deck is LS-DYNA keyword text in a FreeCAD text object, written with the Keyword Editor, templates or by hand, so every keyword the solver's LS-DYNA reader supports can be used (contact, plasticity, rigid walls, initial velocities, ...). The *Run OpenRadioss* dialog runs it in the background with a live log.

Both run the same three programs, use the same solver settings and load the result the same way. They also combine: the FEM solver's `AdditionalKeywords` property appends the keywords of a text object to the generated deck.

4 Workflow 1: OpenRadioss as a FEM workbench solver

This is the quickest way in for FreeCAD FEM users. Set up an analysis as for CalculiX, then choose **Solve** → **Solver OpenRadioss** and run it with **Run solver calculations** (or by double-clicking the solver), as in the upper path of Figure 2.

The last frame is loaded as a regular FEM mechanical result (displacements, von Mises and principal stresses, like CalculiX results) plus a post-processing pipeline. All frames stay in the working directory as `radioss.pvd` for ParaView.

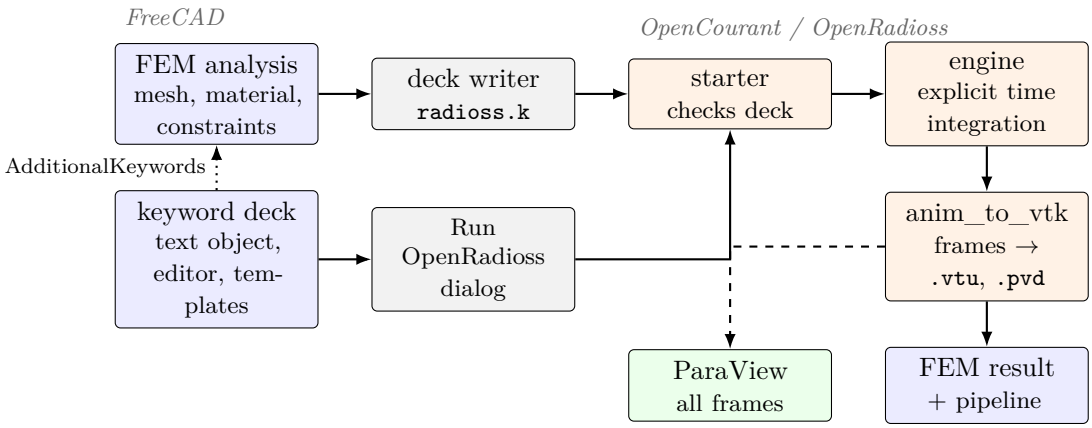


Figure 2: The two ways in. Top: FEM workbench (solver framework, deck writer). Bottom: keyword deck (Run OpenRadioss dialog). Blue: FreeCAD objects, orange: solver programs.

Solver properties. OpenRadioss is an explicit dynamic solver; the properties of the solver object control the time integration (Table 1).

Property	Meaning
`EndTime`	Simulated time (default 0.01 s). Explicit time steps are tiny, so keep it short.
`LoadCurve`	*Ramp*: loads grow linearly from 0 over `EndTime` (quasi-static); *Constant*: loads act fully from the start (impact-like).
`OutputFrames`	Number of result frames written.
`MassScalingMicroseconds`	0 = off; otherwise mass is added so that the time step stays at least this value (e.g. 0.2 µs). Much faster, but check the added mass in the listing.
`Threads`	Threads for starter and engine.
`AdditionalKeywords`	A text document whose keywords are appended to the deck, e.g. from the Keyword Editor.

Table 1: Properties of the *SolverOpenRadioss* object.

What is translated. Volume meshes (tetra4, tetra10, hexa8, penta6, pyra5; second-order hexa, penta and pyramid elements are reduced to their corner nodes) and face meshes with a shell thickness; linear elastic materials, several via references; fixed and displacement constraints (fixed components), force, pressure and self weight. Other constraints are reported and ignored. The deck uses consistent mm / t / s / N / MPa units.

5 Workflow 2: the OpenRadioss workbench (keyword decks)

For models beyond the FEM translation (contacts, other materials, initial velocities, ...) the *OpenRadioss* workbench works directly on LS-DYNA keyword text:

Create Basic .k File adds a small runnable deck (one shell element) as a text object.

Keyword Editor browses 219 keywords with their card layouts, field help and links to the Radioss documentation, fills in card fields and caches the generated keywords; *Update .k File* writes them into the document. A *Templates* menu starts from ready-made keyword sets.

Template Editor creates, edits, reorders and deletes templates with a live preview. User templates are stored in `<FreeCAD user data>/ORfemn/templates.json`.

Run OpenRadioss runs starter, engine and the VTK conversion in the background with a live log and a Stop button, then loads the result into FEM post-processing.

Export LS-DYNA .k File saves a keyword text object to disk.

Both workflows meet in the `AdditionalKeywords` property: keywords written in the editor can extend a deck generated from a FEM analysis.

6 Examples

Three models in `examples/` solve in a few seconds each (Table 2); `examples/README.md` explains how to run them. A rough, AI-generated analysis of their results and meshes is in `docs/ORfe_Example_Analysis.pdf` (not verified by a human).

File	Way in	Result
`cantilever_beam`	FEM, hexahedra, ramp	tip deflection 0.187 mm (beam theory 0.192 mm)
`plate_sudden_pressure`	FEM, shells, sudden load	oscillation with period 2.2 ms (theory 2.24 ms) and peaks of 2× the static deflection
`taylor_impact`	keyword deck	steel bar at 200 m/s on a rigid wall: shortens by 20 %, plastic strain up to 0.81

Table 2: Example models.

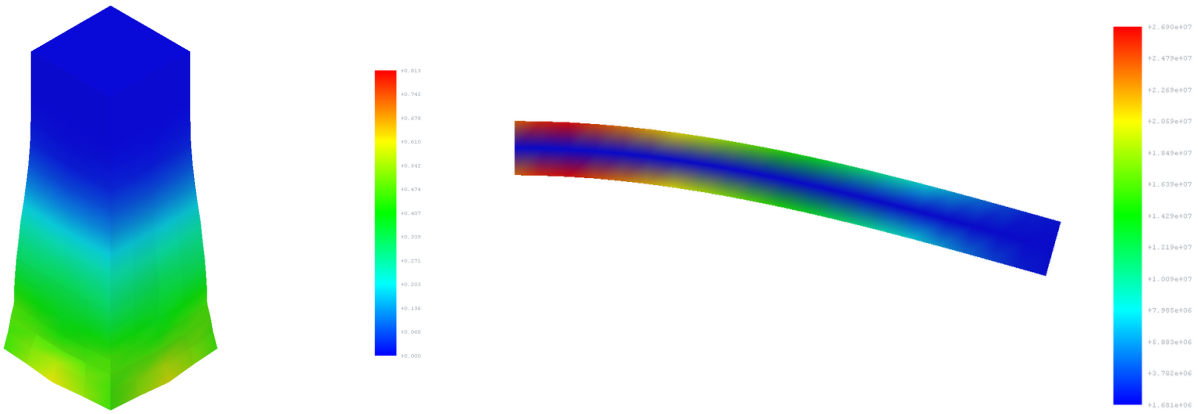


Figure 3: Taylor impact (equivalent plastic strain) and cantilever (von Mises stress, deformation ×100).

7 Testing

The workbench was tested on Debian 13 with FreeCAD 1.0.0 (Debian package), FreeCAD 1.1.4 (Flatpak) and OpenCourant `latest-20261011`:

- 54 unit and integration tests (`python tests/tests.py`): `.k` formatting, templates, deck writer, result conversion, solver detection, requirements check, release download and unpacking, and three real solver runs. All pass.
- Headless FreeCAD runs: a Gmsh cantilever analysis is solved through the FEM solver framework in FreeCAD 1.0 and 1.1, and the document is saved and restored with the solver.
- A GUI test (`tests/freecad_gui_test.py`, started by `tests/run_gui_test.sh`) drives FreeCAD like a user: the preference page with *Test installation*, *Run solver calculations*

on the two FEM examples, the *Run OpenRadioss* dialog with *Load Results in FEM* on the Taylor impact, and the editors. A recording of it is in `docs/video/`, screenshots are in `docs/screenshots/`.

Testing found and fixed three problems: FreeCAD 1.1 changed the arguments of `ObjectsFem.makePostVtkResult`; in the GUI, building the result pipeline from the solver framework's worker thread deadlocked FreeCAD (it is now built in the main thread); and results with a rigid wall failed to load, because the wall's display points have no node number.

Accuracy check. Table 3 compares the tip deflection of a steel cantilever ($100 \times 10 \times 10$ mm, $E = 210\,000$ MPa, $F = 100$ N, ramp load over 10 ms) with beam theory (0.192 mm including shear) and with CalculiX on the same kind of mesh. Hexahedra and linear tetrahedra agree with the reference solutions. With the solver's 10-node tetrahedron the beam deflects about 35 % more than with CalculiX, and the starter reports only 0.70 of the physical mass for every tetra10 mesh tried. The deck's tetra10 node order was checked (every other midside order is rejected by the starter), so this behaviour comes from the solver element, not from the translation. Until this is understood, use hexahedra or a fine tetra4 mesh for quantitative results.

Mesh	Element	OpenCourant [mm]	CalculiX [mm]	Difference
$40 \times 10 \times 10$, $8 \times 2 \times 2$	hexa8	0.01224	–	+0.4 % vs. beam
Gmsh, 4 mm	tetra4	0.1329	0.1307	+1.7 %
Gmsh, 2 mm	tetra4	0.1827	0.1679	+8.8 %
Gmsh, 4 mm	tetra10	0.2592	0.1908	+36 %

Table 3: Tip deflection of the cantilever. Linear tetrahedra are too stiff in both codes (shear locking), so they are compared with CalculiX, not with beam theory. The tetra10 value was computed without mass scaling.

8 Limitations and tips

- Explicit means *dynamic*: a ramp over several periods of the lowest eigenmode gives a quasi-static answer; a constant load gives an impact response with oscillations.
- Mass scaling speeds runs up a lot, but with quadratic tetrahedra it can easily double the mass; check *MASS ADDED* in `radioss_0001.out`.
- Only linear elastic materials and the constraints listed above are translated; anything else can be added as keywords through `AdditionalKeywords`.
- For bending with hexahedra use at least two elements through the thickness.
- Tested on Linux; the code also handles Windows and arm64 executable names, but these platforms are untested. OpenCourant has no macOS release.

9 Source, license and feedback

Source code, issue tracker and documents:
https://github.com/CalzitPLA/FC_OpenCourant.
ORfe is free software under the GNU Lesser General Public License, version 2.1 or later, like FreeCAD itself. OpenRadioss and OpenCourant are separate programs under the AGPL; ORfe only runs their executables. Feedback is welcome on the FreeCAD forum (FEM section) or as GitHub issues.

No warranty. The workbench, the examples and this document are provided "as is", without warranty of any kind; simulation results must be checked by a qualified engineer before they are relied on. This introduction was drafted with the help of an AI assistant from the test runs described above; the separate example analysis is entirely AI-generated and not verified by a human.